

	Shri Gurbh Singhji Institute of Engineering and Technology, Nanded Department of BSH&M Summer Term Examination Academic Year: 2024-25	
Class: FY B.Tech.	Subject: Professional Communication	Subject Code: AES102/HMC110
Date: 25.06.2025	Time: 10:00am to 01:00pm	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange Sharing of stationery, calculator etc. not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Attempt all of the following questions.	Marks	CO	BT
Q.1	A Do as directed. (Any five)	10	CO1	BT1
	1 I have to choose ... the two prizes. (between/among/amongst)			
	2 I would travel by taxi. (Write the sentence structure)			
	3 He bought ... pen ... and ink ... for ... union. (Use articles)			
	4 When will he learn these things? (Change into passive voice)			
	5 I said, "Let's go to cinema." (Change into indirect narration)			
	6 Scientists have discovered medicine for COVID. (Change into simple present tense.)			
	B Correct the following sentences by rewriting it. (Any five)	10	CO1	BT1
	1 I am having four shirts and three pants.			
	2 Ask her to sit besides me.			
	3 Did you went to the park?			
	4 He could not decide among the two choices.			
	5 My younger brother is good in mathematics.			
	6 Monkeys are climbing about tree.			
Q.2	A United India Ltd. is looking for young, dynamic and result oriented engineers from all branches. Write an application with CV to the CEO, United India Ltd., Bangalore.	10	CO2	BT2
	B Develop a well-knit paragraph on any one of the topics: 'Beat plastic pollution' OR 'Give blood, give hope'	10	CO2	BT3
Q.3	Write an order for 60 computers from Microbe Technologies, Chennai by providing all details of the quotation you received.	10	CO3	BT3
Q.4	As a CR of the class, write a report focusing the causes related to the high absenteeism of the students to the class.	10	CO3	BT4
	Assume that you are the head of the Department of Mechanical Engineering. You need some new and updated equipment for your mechanical lab. Write a letter of enquiry to Moto-Pro Ltd., Indore giving full details of your requirements in complete block style.	10	CO4	BT4
Q.6	A Suman Textiles Co. Ltd. Hyderabad has been facing a decline in profits by various reasons as fall in sales. As a marketing manager, you have been asked by the Managing Director to investigate into causes and make suitable recommendations as a report.	10	CO3	BT6
	B Write a precis of the following paragraph with suitable title.	10	CO4	BT5

India may be making rapid progress in IT industry and in primary education, but with the absence of value-based education they seem to have neither the motivation nor the wisdom to move in a joyous direction. These values are not readily available through the television either. Religion, which should have been the source of imbibing values has become dangerously ritualistic. Moreover, all religions actively advance irrelevant logic against other religions. Books can provide joyous wisdom. It is surprising that no one seems to be actively seeking it through the printed media. Almost all Indians have lost the pleasure of reading the languages. In fact, they do not even seem to be aware that there is an endless treasure waiting to be tapped through the books in Indian languages. It is up to the intellectuals to provide leadership and inspiration to those who cannot master the necessary motivation to read books in Indian languages

Q.8 As a Lab in charge write a letter of complaint to M/S Das & Co. for the delivery of defective equipment for your laboratory costs about 2 Lakh asking them about replacement.

10 CO4 BT6

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

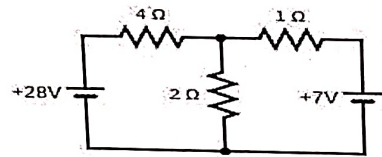
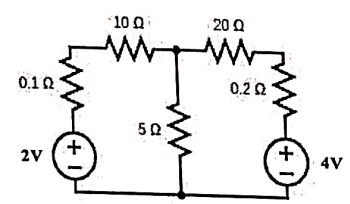
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To enable students to learn the structure and style of effective sentences.
CO 2	To enable students to express themselves in English effectively.
CO 3	To enhance students' Listening, Speaking, Reading & Writing Skills.
CO 4	To help students to understand nuances of technical writing.

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering Summer Term Examination Academic Year: 2024-25		
	Class: First Year	Subject: Basic Electrical Engineering	Subject Code: ESC 101/105
	Date: 24/06/2025	Time: 10:00 to 13:00	Max. Marks: 100

Note:

- All questions are compulsory
- Step-wise marks are given for numerical questions.
- A scientific calculator is allowed.
- This paper contains 02 pages (back to back)

Q.N.	All questions are compulsory			Marks	CO	BT
Q1	a)	i) State and explain ohms law and what are its limitations ii) State and explain Kirchhoff's both laws.		6	CO-1	1
	b)	What are the types of electrical sources? Explain Independent and Dependent sources with symbols.		6	CO-1	1
	c)	Define i) Current, ii) Voltage , iii) Electrical Power, iv) Resistance		4	CO-1	1
Q.2	a)	State Superposition theorem and write steps to solve the Superposition theorem		7	Co-2	2
	b)	Write steps to solve mesh Analysis and Node analysis with a circuit diagram		7	CO-2	2
	c)	Find the current through 2Ω resistor using Thevenin's theorem. 		7	CO-2	3
	d)	Find the current through 5Ω resistor using Norton's theorem. 		7	CO-2	3
Q.3	a)	Compare Electric circuits and Magnetic circuits		8	CO-3	1
	b)	State Faraday's laws of electromagnetic induction. (Both the Laws		4	CO-3	1
	c)	State Fleming's left and Right-hand Rule		4	CO-3	1
	d)	An iron ring with a mean circumference of 0.8 meters is uniformly wound with 550 wire turns. Calculate the flux density value that a current of 1.1 Amp would produce in the ring. (Assume $\mu_r = 1400$)		5	CO-3	3
Q.4	a)	Explain the advantages of a phase system over a single-phase system.		6	CO-4	2

	b)	Define i) Cycle ii) RMS Value iii) Instantaneous Value iv) Frequency	4	CO-4	
	c)	A Resistance of $5\ \Omega$ and Capacitive reactance of $10\ \Omega$ are connected in series if the current through the circuit is 3A; find impedance, phase angle, and power factor.	5	CO-4	3
Q.5	a)	Draw and explain the construction of single-phase transformers and their application	6	CO-5	2
	b)	Explain Open circuit (OC) and Short circuit (SC) test of 1 phase transformer	7	CO-5	2
	c)	Draw and Explain the working principle of a Permanent Magnet Moving Coil (PMMC) meter with a neat diagram.	7	CO-5	2

————— ALL THE BEST —————

Bloom's Taxonomy (BT):-

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:				
CO 1	Describe the basic concepts of voltage, current, EMF, energy sources, and fundamental laws like Ohm's and Kirchhoff's in electric circuits.				
CO 2	Analyze DC electrical circuits using series-parallel combinations, mesh and nodal analysis, and theorems such as Superposition, Thevenin's, and Norton's.				
CO 3	Apply electromagnetic laws to analyse magnetic circuits, induced EMF, and energy storage in magnetic fields				
CO 4	Analyze single-phase and three-phase AC circuits using phasor diagrams, and calculate impedance, power factor, and power components.				
CO 5	Explain the construction and operation of single-phase transformers and basic electrical measuring instruments like PMMC meters and wattmeters.				

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded		
Class: FY B.Tech.		Subject: Mathematics-II (Multivariable Calculus, Partial Differential Equations, Laplace Transforms)	
Date: 17/06/2025		Time: 10.00-13.00	
		Subject Code: BSC104	
		Max. Marks: 100	

Note: i) calculators are not allowed. ii) This paper contains total 2 pages.

Q.1	Attempt any five	marks	CO	BT
a	Find the partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ & $\frac{\partial^2 f}{\partial x \partial y}$ for the function $f(x, y) = \ln(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$.	06	CO1	BT1
b	Express $\frac{\partial z}{\partial u}$ & $\frac{\partial z}{\partial v}$ as a function of u & v by using the Chain Rule, if $Z = \tan^{-1}\left(\frac{x}{y}\right)$, $x = u \cos(v)$, $y = u \sin(v)$.	06	CO1	BT2
c	If $f(x, y) = \frac{x-y}{xy+2}$ then find i) The directions in which $f(x, y)$ increases and decreases most rapidly at $p_0(1, -1)$. ii) The directional derivative of $f(x, y)$ in the direction of $\vec{u} = 2i - j$ at $p_0(1, -1)$.	06	CO1	BT2
d	Find equation of tangent plane and normal line to the surface $x \ln(y) + 2y \ln(z) - x = 0$ at the point $p_0(2, 1, e)$.	06	CO1	BT2
e	Find all the local maxima, local minima & saddle points (if any) of the function $f(x, y) = 6x^2 - 2x^3 + 3y^2 + 6xy$.	06	CO1	BT3
f	Find solution of following IVP by using Laplace transform. $y'' + y' - 6y = 0$, $y(0) = 0$, $y'(0) = 1$.	06	CO4	BT3
Q.2	Attempt any five			
a	Evaluate the integral $\int_0^1 \int_0^1 (xy)^2 \sin(\pi x^3) dx dy$.	06	CO1	BT2
b	Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^\pi \int_x^\pi \frac{\sin y}{y} dy dx$.	06	CO1	BT2
c	Evaluate the following integration over the given box B. $\iiint_B ze^{x+y} dx dy dz$, $B = [0, 1] \times [0, 1] \times [0, 1]$.	06	CO1	BT2

	d	Let curve C is $c(t) = (\sin t, \cos t, t)$, with $0 \leq t \leq \pi$ and $F(x, y, z) = xi + yj + zk$ compute $\int_C F \cdot ds$.			
	c	Find the volume of the region bounded above by the paraboloid $z = x^2 + y^2$ and below by the triangle enclosed by the lines $y = x$, $x = 0$ & $x + y = 2$ in the xy -plane.	06	CO1	BT2
	f	Find the inverse Laplace transform of following functions: i) $F(s) = \frac{1}{s^2 - 5s + 6}$ ii) $F(s) = \frac{6s+7}{s^2 + 2s + 5}$	06	CO1	BT2
Q.4 Attempt all					
	a	Find the extreme values of $f(x, y) = x^2 - y^2$ on the circle C of radius 1 centred at the origin in the xy -plane.	06	CO1	BT3
	b	Compute divergence and curl of the vector field $F(x, y, z) = 2xye^z \mathbf{i} + e^z x^2 \mathbf{j} + (x^2 ye^z + z^2) \mathbf{k}$.	06	CO1	BT1
	c	Find the work done by the force field $F(x, y) = e^{-y} \mathbf{i} - xe^{-y} \mathbf{j}$ in moving an object along the path $c(t) = (2 - 2t, t)$, $0 \leq t \leq 1$.	06	CO1	BT3
	d	Find the Fourier cosine series to represent, $f(x) = x^2 + 2$, $-\pi \leq x \leq \pi$ & $f(x + 2\pi) = f(x)$.	06	CO2	BT2
	e	Find the Laplace transform of following functions: i) $f(t) = \sin(2t) \cdot \cos(3t)$ ii) $f(t) = t^3 e^{-2t}$ iii) $f(t) = e^{-t} \sin^2 t$.	06	CO4	BT2
Q.3 Attempt all					
	a	Find the Fourier series of the periodic function $f(x)$ defined as $f(x) = \begin{cases} 2, & -\pi \leq x < 0 \\ x, & 0 \leq x < \pi \end{cases}$ & $f(x + 2\pi) = f(x)$.	08	CO2	BT2
	b	Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{-x}{\sqrt{x^2 + y^2}}$ does not exist. OR Show that the vector field $V(x, y, z) = xi + yj + zk$ cannot be the curl of some vector field F .	02	CO1	BT2

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]

CO1. Apply the knowledge of multivariable calculus to solve problems related to data Science, optimization, electrical, mechanical, chemical process.

CO2. Apply the concept of Fourier series and their application in system communication and digital signal processing, vibrations.

CO3. Examine the concept of partial differential equations to describe the heat and wave propagation phenomenon arising in fluid mechanics, electromagnetic theory, mechanical, and other engineering problems.

CO4. Use the knowledge of Laplace Transforms of customary functions arising in network analysis and control systems and solve ODE's and PDEs optimization electrical, mechanical, chemical process.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded**

Department of BSH&M (Mathematics)

Summer term Examination,

Academic Year: 2024-25 (Semester-I)

Class: FY B.Tech.	Subject: Mathematics-I (Calculus and linear algebra)	Subject Code: BSC103
Date: 18.06.2025	Time: 10:00 to 13:00	Max. Marks: 100

Note: i) calculators are not allowed. ii) This paper contains total two pages

Q.1	Attempt any five	Marks	CO	BT
a	Verify that the following function satisfies the three hypotheses of Rolle's Theorem on the given interval. Find all numbers "c" such that f satisfy the conclusion of Rolle's Theorem. $f(x) = \sqrt{x} - \frac{1}{3}x$, $[0,9]$.	06	CO1	BT2
b	Find the absolute maximum and absolute minimum values of the function on the given interval $f(x) = 3x^2 - 12x + 5$, $[0,3]$.	06	CO1	BT3
c	Find: (a) $\lim_{x \rightarrow \infty} \frac{e^x}{x^2}$ (b) $\lim_{x \rightarrow 0} \frac{\sin x - x}{x^3}$	06	CO1	BT2
d	Find Taylor's series of $f(x) = (1+x)^{-2}$ about $a = 0$ (ie find Maclaurin series).	06	CO1	BT2
e	i) If $f(1) = 10$ and $f'(x) \geq 2$ for $1 \leq x \leq 4$, how small can $f(4)$ be possible? ii) Use the mean value theorem to prove the inequality $ \cos(a) - \cos(b) \leq a - b , \forall a, b \in \mathbb{R}$.	06	CO1	BT3
f	Solve the following differential equation $\frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} = 2xe^{-x}$.	06	CO3	BT2
Q.2	Attempt any five			
a	Evaluate the upper and lower sums for $f(x) = 1 + x^2$, $-1 \leq x \leq 1$, with $n = 4$.	06	CO2	BT1
b	i) Use Fundamental theorem of calculus to find $f'(x)$ for the function $f(x) = \int_1^x \frac{1}{t^3+1} dt$. ii) Check the following differential equation is exact or not? $(2xy + e^y)dx + (x^2 + xe^y)dy = 0$	06	CO3	BT2
c	Find rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 1 & 2 & 1 \\ 3 & -1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{bmatrix}$.	06	CO3	BT2

	d	Find all eigenvalues of the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix}$.	06	CO3	BT2
	e	Sketch the region enclosed by the following parabolic curves & find its area: $y = x^2$, $y = -x^2 + 2$.	06	CO2	BT3
	f	Find the volume of the solid obtained by rotating the region bounded by the curves $y = x^2$, $y = 0$, $x = 0$ & $x = 1$ about the x -axis.	06	CO2	BT3
Q.3 Attempt all					
	a	If $f(x) = 2x^3 + 3x^2 - 36x$ then, (i) Find the interval on which f is increasing or decreasing. (ii) Find the local maximum and local minimum values of f . (iii) Find the interval of concavity and the inflection points.	10	CO1	BT3
	b	For what values of λ and μ the system of linear equations: $x + y + z = 6$ $x + 2y + 5z = 10$ $2x + 3y + \lambda z = \mu$ has (i) a unique solution (ii) no solution (iii) infinite solutions (iv) Find the solution for $\lambda = 2$ and $\mu = 8$.	10	CO4	BT3
	c	Find all eigenvectors associated to eigenvalues $\lambda_1 = 1$ & $\lambda_2 = 7$ of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$. (Do not compute eigenvalues, take eigenvalues $\lambda_1 = 1$ & $\lambda_2 = 7$ for matrix A)	10	CO4	BT2
	d	i) Solve the following differential equation by variation of parameters $\frac{d^2y}{dx^2} - y = \frac{2}{1+e^x}$. [you can directly use $\int \frac{1}{e^x(1+e^x)} dx = -e^{-x} + \log(e^{-x} + 1)$]	08	CO3	BT2
		ii) Find the order and degree of the following differential equation $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 8y = \tan x$.	02		

[BT1: Remembering, BT2: Understanding, BT3: Applying, BT4: Analysing, BT5: Evaluating, BT6: Creating]
CO1. Understand the concepts of derivatives and apply the knowledge of differentiation to solve problems related to polar curves, radius of curvature and related engineering problems.
CO2. Learn the definite and improper integrals and apply the notion to engineering problems.
CO3. Learn the notion of ordinary differential equations and use them in electric circuit problem, communication, optimization, etc.
CO4. Make use of matrix theory for solving of system of linear equations and compute eigen values and eigenvectors and solve the problems related to his engineering field in data science, electrical, mechanical, chemical process, optimization, etc.